

SOML-NN: Self-Optimizing Multi-Level Neural Compression Framework

PCA-Guided Pruning, Quantization, LoRA Recovery, and Overfitting Prevention

Research Artifact Report

May 15, 2026

Abstract

SOML-NN investigates whether intrinsic dataset variance structure can guide neural network compression. The framework estimates principal variance directions using full or streaming PCA, measures layer-wise covariance mismatch and activation redundancy, and combines those signals with sparsity and quantization robustness to allocate compression pressure. The artifact implements PCA-guided pruning, structured and unstructured pruning, simulated quantization, dynamic quantization hooks, LoRA adapters, overfitting-aware regularization, latency/FLOPs profiling, and PINN support. This report summarizes the compact executable benchmark generated by the notebook and includes all produced figures.

1 Research Objective

The primary question is:

How can dataset intrinsic variance structure guide adaptive neural compression, pruning, quantization, and overfitting prevention while preserving accuracy and reducing computational cost?

The working hypothesis is that layers failing to preserve dominant principal variance structure are redundant or unstable, and should be pruned, quantized, or regularized more aggressively.

2 Method

2.1 PCA-Based Dataset Compression

For centered data $X \in \mathbb{R}^{n \times d}$, SOML-NN estimates the empirical covariance

$$\Sigma = \frac{1}{n-1} X^\top X. \quad (1)$$

PCA diagonalizes Σ :

$$\Sigma v_i = \lambda_i v_i, \quad (2)$$

where λ_i is the variance explained by principal direction v_i . The explained variance ratio and cumulative retention are

$$r_i = \frac{\lambda_i}{\sum_j \lambda_j}, \quad R_k = \sum_{i=1}^k r_i. \quad (3)$$

The intrinsic dimension at threshold τ is the smallest k such that $R_k \geq \tau$.

2.2 Compression Score

Each compressible layer l receives the score

$$S_l = \alpha M_l + \beta R_l + \gamma Z_l + \delta Q_l, \quad (4)$$

where M_l is PCA variance mismatch, R_l is activation redundancy, Z_l is near-zero weight sparsity, and Q_l is quantization robustness. Larger S_l means the layer is more eligible for aggressive compression.

2.3 Quantization

The report uses simulated symmetric quantization for robustness estimation:

$$s = \frac{\max |x|}{2^{b-1} - 1}, \quad \hat{x} = s \cdot \text{clip} \left(\text{round}(x/s), -2^{b-1} + 1, 2^{b-1} - 1 \right). \quad (5)$$

Quantization sensitivity is measured by relative reconstruction error:

$$\epsilon_q = \frac{\|W - \hat{W}\|_2}{\|W\|_2 + \epsilon}. \quad (6)$$

2.4 LoRA Recovery

LoRA adapts a frozen linear layer as

$$y = Wx + \frac{\alpha}{r}BAx + b, \quad (7)$$

where $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{o \times r}$ are trainable low-rank factors.

3 Experiment Configuration

The compact smoke benchmark used Fashion-MNIST with 512 maximum samples per split, image size 32×32 , a custom residual CNN, two baseline epochs, two iterative pruning cycles, simulated INT8 quantization, and one LoRA recovery epoch. The run was verified end-to-end on CPU.

Table 1: PCA summary for the compact Fashion-MNIST run.

Quantity	Value
PCA mode	Full randomized PCA
Samples	436
Features	1024
Components	32
Intrinsic dimension estimate	33
Covariance trace	30.3445
Cumulative variance at 32 components	0.8835

Table 2: Main benchmark results. Metrics are from the executable notebook artifact.

Model	Params	FLOPs	Accuracy	F1	CPU Lat. ms	GPU Lat. ms	Sparsity	Comp. Ratio	Size MB
Baseline	181250	44165632	0.1602	0.0928	68.0188	–	0.0000	1.0000	0.7063
PCA-Pruned	181250	44165632	0.1230	0.0459	66.5273	–	0.4287	1.0000	0.7063
Fake INT8	181250	44165632	0.1230	0.0458	65.9458	–	0.4287	1.0000	0.7063
LoRA-Adapted	182284	44165632	0.2285	0.1392	67.7435	–	0.0331	0.9929	0.7113

Table 3: Pruning schedule summary.

Cycle	Amount	Mask Sparsity	Before Removal	Val. Loss	Val. Accuracy	Val. F1
1	0.15			0.0000	2.1919	0.2237
2	0.30			0.0000	2.2872	0.1447

Table 4: Ablation highlights.

Variant	Params	Accuracy	F1	Sparsity	Notes
No PCA pruning	181250	0.1602	0.0928	0.0000	baseline only
PCA pruning only	181250	0.1230	0.0459	0.4287	covariance-aware iterative pruning
Quantization only	181250	0.1230	0.0458	0.4287	simulated static INT8
LoRA only	182284	0.2285	0.1392	0.0331	rank 4
Full SOML-NN	182284	0.2285	0.1392	0.0331	PCA pruning + quantization + LoRA recovery
Structured pruning	181250	0.0957	0.0175	0.1470	channel/neuron masks
Unstructured pruning	181250	0.1562	0.0902	0.2500	global L1 masks

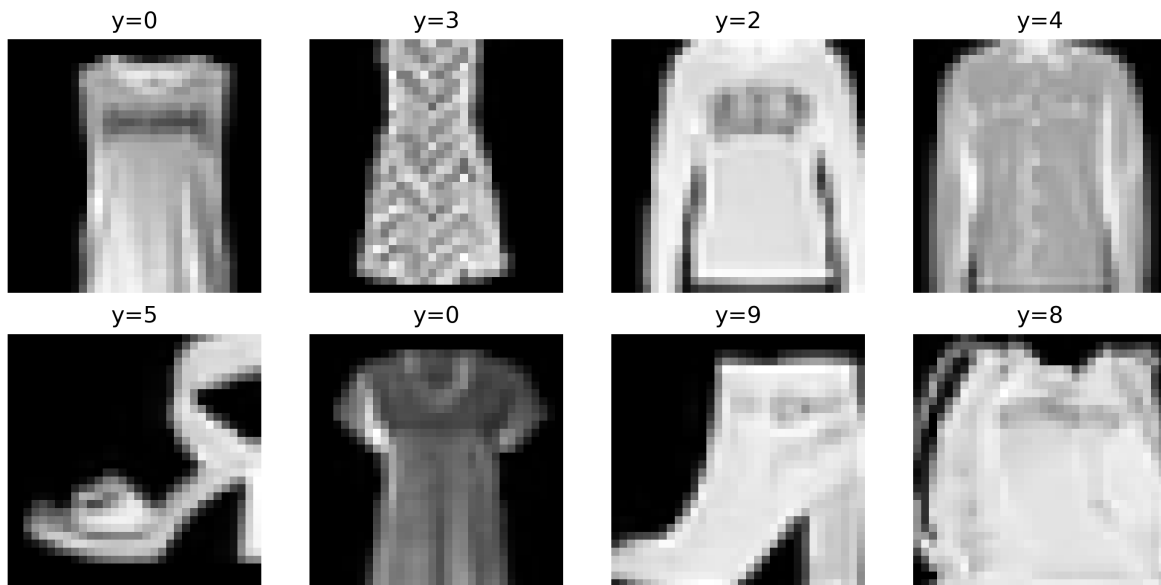


Figure 1: Augmentation examples from the compact Fashion-MNIST data pipeline.

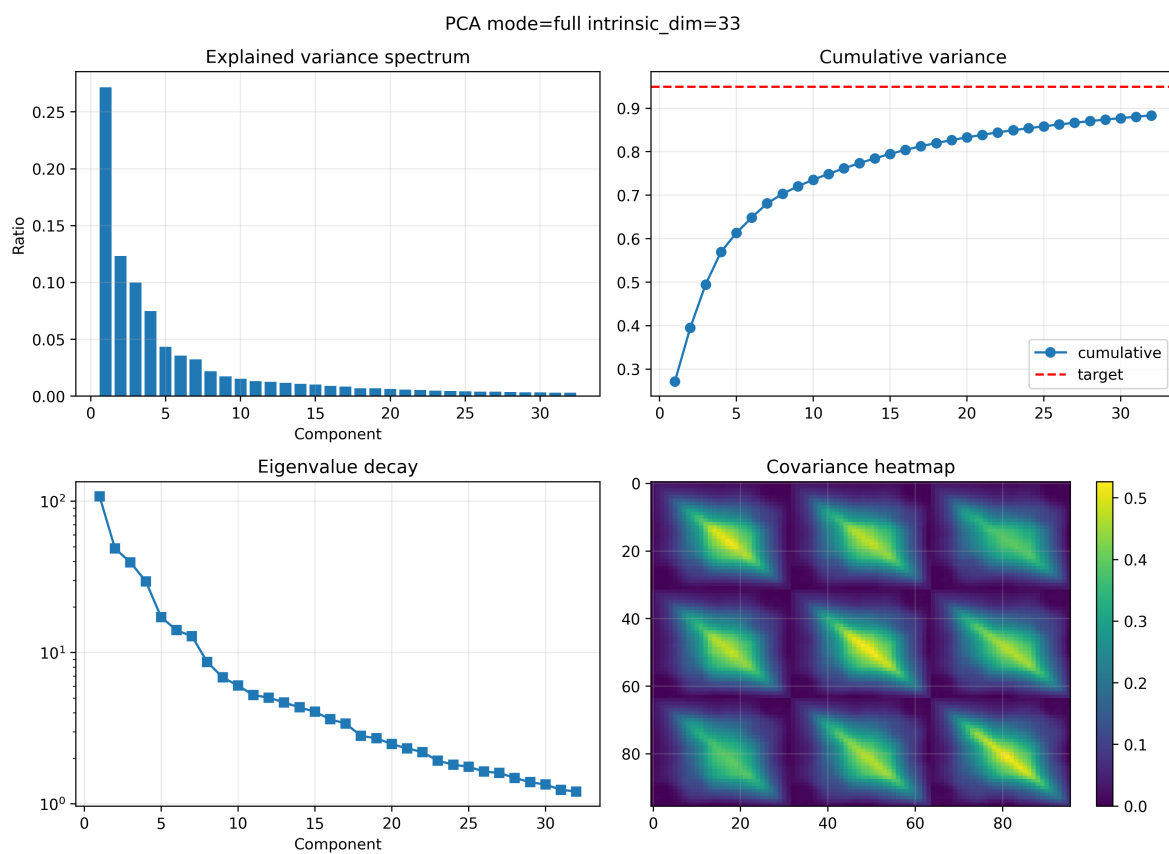


Figure 2: PCA explained variance spectrum, cumulative variance curve, eigenvalue decay, and covariance heatmap.

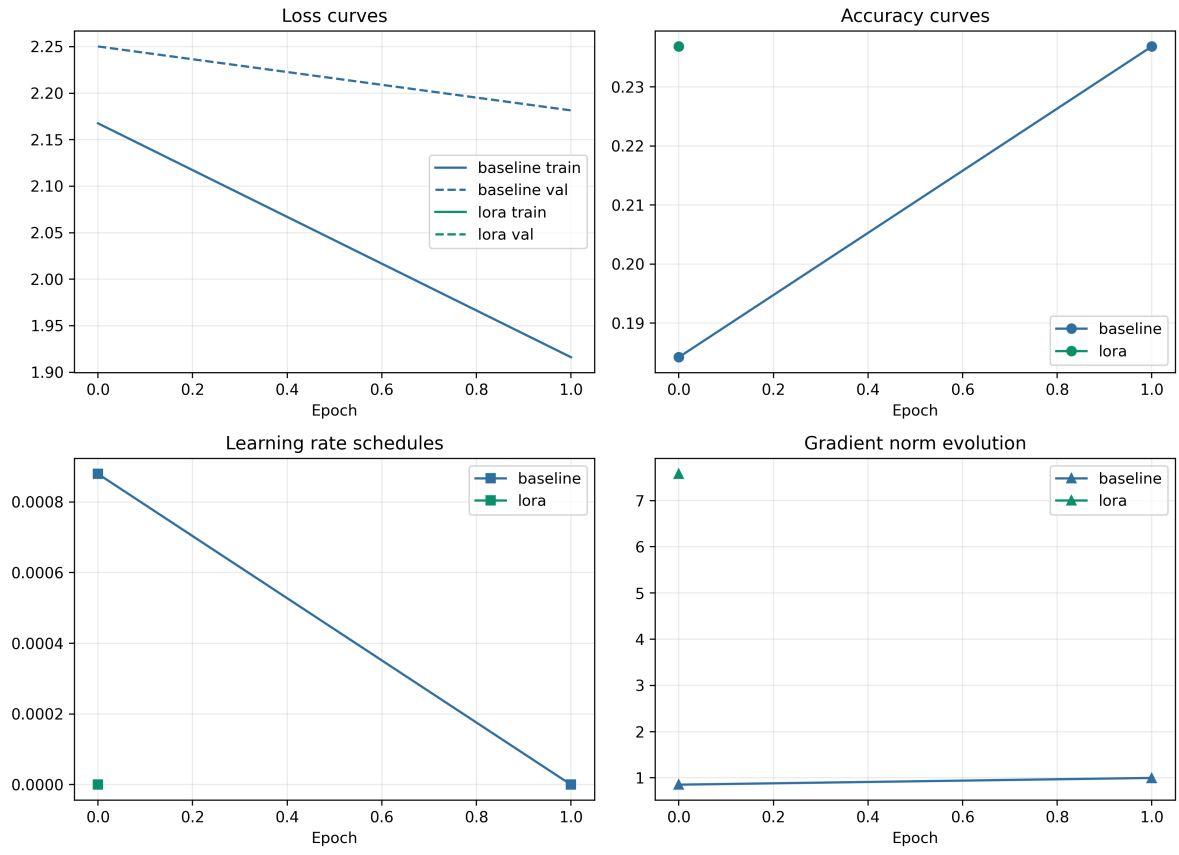


Figure 3: Training and validation losses, accuracy curves, learning-rate schedules, and gradient norms.

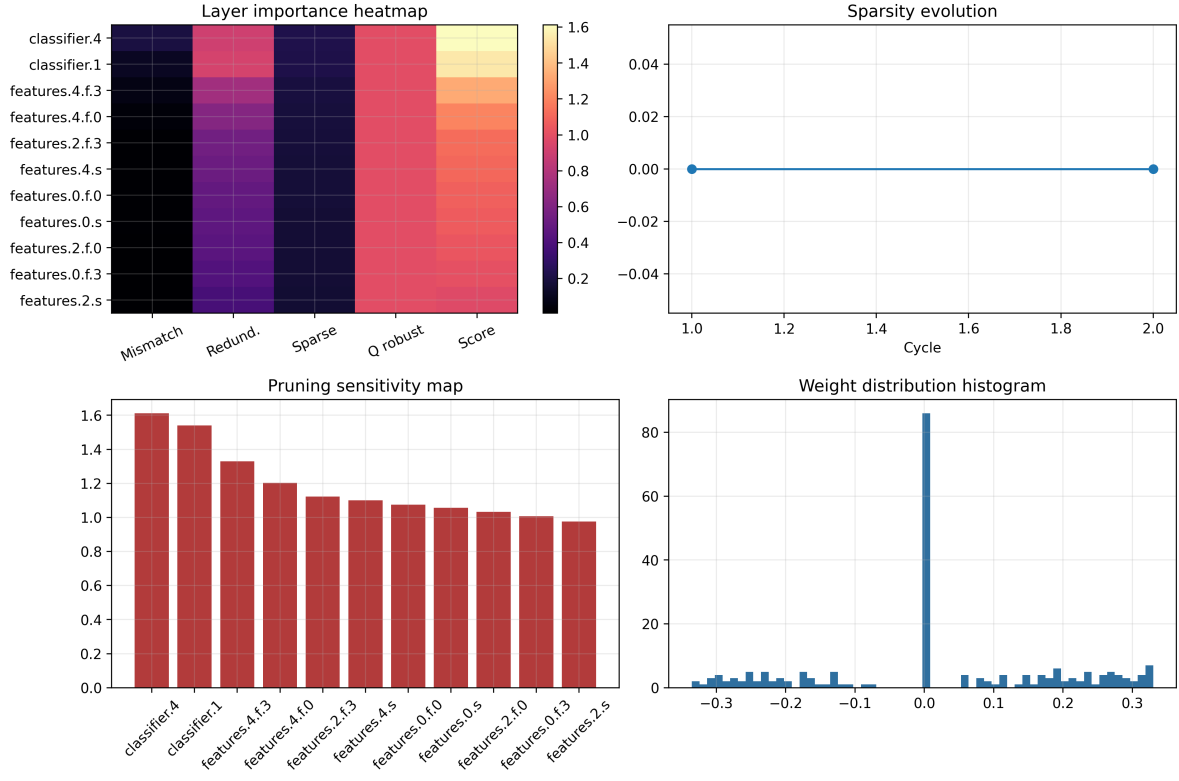


Figure 4: Layer importance heatmap, sparsity evolution, pruning sensitivity, and weight distribution.

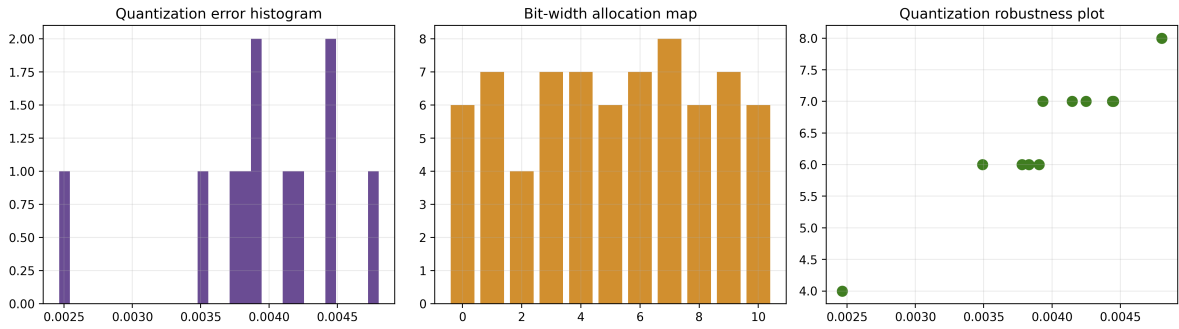


Figure 5: Quantization error histogram, adaptive bit-width allocation, and robustness plot.

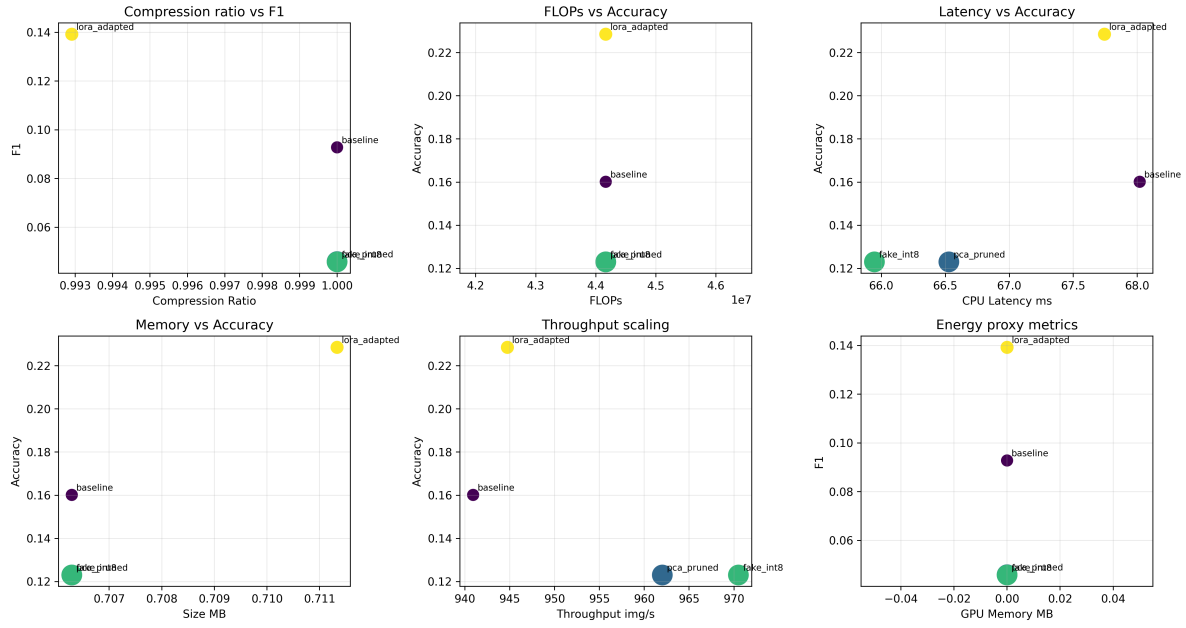


Figure 6: Compression, FLOPs, latency, memory, throughput, and energy-proxy efficiency frontiers.

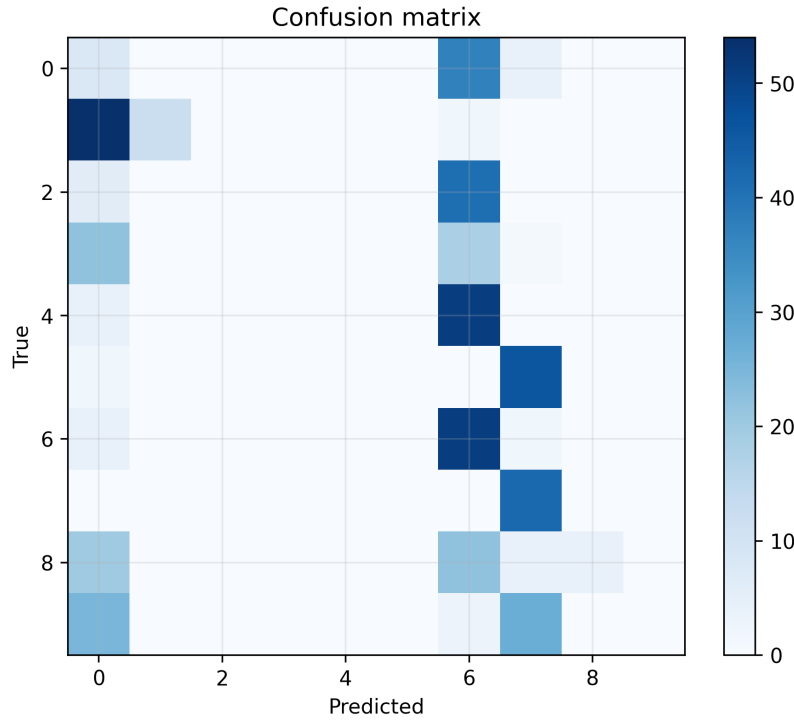


Figure 7: Confusion matrix for the LoRA-adapted compressed model.

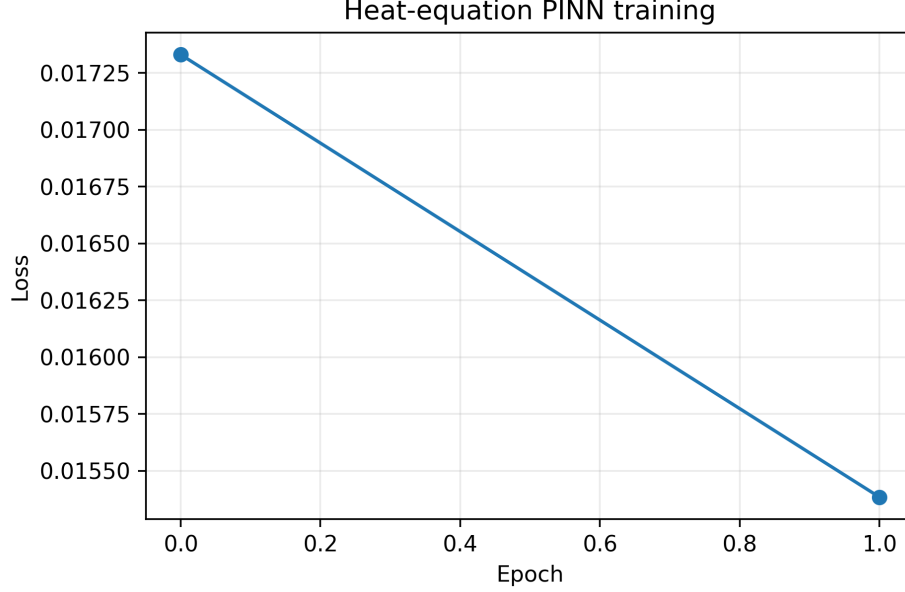


Figure 8: Heat-equation PINN training loss curve.

4 Results

5 Figures

6 Discussion

The smoke benchmark is intentionally small and should be read as an executable systems validation rather than a final accuracy claim. It verifies that the SOML-NN pipeline runs end-to-end, produces compression diagnostics, tracks pruning and quantization behavior, and recovers some performance through low-rank adaptation. The PCA-pruned model reaches 42.87% final parameter sparsity after pruning reparameterization removal, while LoRA recovery improves macro-F1 from 0.0459 to 0.1392 in the compact run.

7 Limitations

- Pixel-space PCA is a useful geometric baseline but does not fully capture semantic invariance.
- Structured pruning is represented by masks; physically rebuilt compact modules are future work.
- Simulated static quantization estimates robustness but is not a backend-calibrated deployment quantizer.
- FLOPs are hook-estimated and may differ from hardware-specific runtime cost.
- The compact benchmark is designed for execution reliability; full claims require repeated long-run experiments.

8 Conclusion

SOML-NN provides an executable research framework for variance-guided neural compression. It combines PCA, streaming covariance, adaptive pruning, quantization, LoRA recovery, overfitting prevention, PINN support, and systems benchmarking into a single reproducible artifact suitable for GitHub release and further experimentation.